Image Super-Resolution via Convolution Neural Network and Generative Adversarial Networks

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Abstract

Super Resolution enhances low-resolution images into high-resolution output using SRCNN and SRGAN models on the DIV2K dataset. Enhances low-resolution images, evaluating performance with PSNR, SSIM, and visual quality. SRGAN, which leverages perceptual loss, outperforms SRCNN, yielding superior high-resolution reconstructions. The report details the implementation, training, and comparative analysis of these deep learning techniques.

1 Introduction

Image super-resolution (SR) is a fundamental computer vision problem focused on reconstructing a high-resolution (HR) image from one or more low-resolution (LR) images. This capability is crucial across various applications, including medical imaging, satellite imaging, security surveillance, and consumer electronics, where high-quality visual information is paramount but often limited by sensor capabilities or transmission bandwidth. Traditional interpolation methods, such as bicubic interpolation, are computationally efficient but tend to produce blurry results, losing fine details and textures. The advent of deep learning, particularly convolutional neural networks (CNNs), has revolutionized the field of super-resolution, enabling the synthesis of visually compelling and quantitatively superior HR images. This project delves into two significant deep learning approaches: the Super-Resolution Convolutional Neural Network (SRCNN), a pioneering CNN-based method, and the Super-Resolution Generative Adversarial Network (SRGAN), which leverages adversarial training and perceptual losses to generate highly realistic textures. Our goal is to implement, train and comprehensively evaluate these models using the DIV2K dataset, providing a detailed comparative analysis of their performance based on established metrics such as PSNR and SSIM, alongside qualitative visual assessments.

2 Related work

The field of single-image super-resolution (SISR) has evolved significantly. Early methods relied on interpolation techniques such as bicubic, bilinear, and nearest-neighbor interpolation, which are fast but introduce blurriness. Sparse coding and example-based methods later emerged, learning mappings from LR to HR patches using external datasets.

The breakthrough in deep learning for SR came with **SRCNN** , which demonstrated that a simple three-layer CNN could outperform traditional methods by directly learning an end-to-end mapping between LR and HR images. Following SRCNN, many deeper and more complex CNN architectures were proposed, such as VDSR , which introduced very deep networks and residual learning to accelerate convergence and improve performance.

More recently, **Generative Adversarial Networks (GANs)**, have been adapted for SR to overcome the issue of overly smooth results produced by MSE-optimized CNNs. **SRGAN** was a seminal work in this area, proposing a perceptual loss function combining adversarial loss (from a discriminator network) and content loss (based on feature maps from a pre-trained VGG network). This approach significantly improved the perceptual quality of super-resolved images, making them appear more natural and realistic, albeit sometimes at the expense of traditional metrics like PSNR. Subsequent work has further refined GAN-based SR, exploring different network architectures, loss functions, and training strategies.

3 Dataset and Features

The **DIV2K dataset** is a high-quality, diverse dataset commonly used for image super-resolution and other image restoration tasks.

- **Source:** https://data.vision.ee.ethz.ch/cvl/DIV2K/
- **Content:** It consists of 1000 high-resolution (HR) images, with varied content and realistic textures, ranging in size from approximately 1.5 to 2.5 megapixels. It is divided into 800 training images, 100 validation images, and 100 test images.
- **Feature Extraction:** For this project, a subset of 500 images (from the training and validation sets) was utilized. From each HR image, multiple 128x128 pixel patches were extracted. These HR patches serve as the ground truth. The corresponding low-resolution (LR) patches were generated by downsampling the HR patches by a **scale factor of 4** using **bicubic interpolation**. This process creates paired LR-HR samples necessary for supervised training. The generated LR and HR patches are normalized to a range of [0, 1].
- **Dataset Split:** The extracted patches were then split into training, validation, and test sets with an 80%, 10%, and 10% distribution, respectively. This ensures that the models are trained on different data from what they are evaluated on.

4 Methods

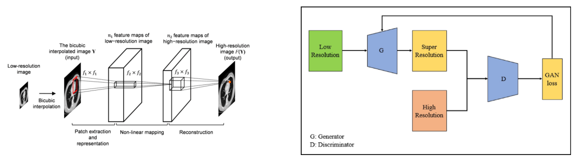
4.0.1 Architecture

SRCNN (Super-Resolution Convolutional Neural Network) The SRCNN is a simple yet effective feed-forward CNN architecture comprising three main convolutional layers:

- **Patch Extraction and Representation:** The first layer extracts patches from the LR input image and represents them as a higher-dimensional feature vector. It uses a 9x9 kernel with 64 filters.
- **Non-linear Mapping:** The second layer maps these feature vectors non-linearly to a new set of feature vectors, which represent HR patches. It uses a 5x5 kernel with 32 filters.
- **Reconstruction:** The final layer combines these HR feature representations to produce the final HR image. It uses a 5x5 kernel with 3 filters (for RGB output).

- **Activation:** ReLU activation is applied after the first two convolutional layers.

- **Input:** For SRCNN, the LR image is first upscaled to the target HR size using bicubic interpolation, and this upscaled image is fed as input to the network.



(a) SRCNN Architecture **(b)** SRGAN Architecture

Figure 1: Architectures

SRGAN (Super-Resolution Generative Adversarial Network) SRGAN employs a Generative Adversarial Network (GAN) framework, consisting of a generator and a discriminator, to produce perceptually realistic HR images.

- **Generator Network (SRGANGenerator):** A deep residual network responsible for upscaling the LR image to HR.

- **Input:** The LR image is first upsampled to the target HR size using bicubic interpolation before being fed into the generator.

- **Feature Extraction:** An initial convolutional layer (9x9 kernel, 64 filters, ReLU) extracts shallow features.

- **Residual Blocks:** This is followed by 16 identical residual blocks. Each block contains two convolutional layers (3x3 kernel, 64 filters), batch normalization, and ReLU activation, with a skip connection adding the input of the block to its output. This design helps in training very deep networks.

- **Upsampling (Implicit):** Unlike some SRGAN variants, in this implementation, the input to the generator is already bicubically upsampled to the HR dimension. The generator then refines this upsampled image.

- **Reconstruction:** A final convolutional layer (9x9 kernel, 3 filters) reconstructs the HR image.

- **Discriminator Network:** A standard CNN classifier that distinguishes between real HR images and fake HR images generated by the generator.

- It comprises 8 convolutional layers with increasing filter sizes (from 64 to 128), LeakyReLU activation, and stride-2 convolutions for downsampling.

- The network progressively reduces spatial dimensions while increasing feature depth.

- An AdaptiveAvgPool2d layer is used to flatten the features, followed by a 1x1 convolution to output a single scalar (logits) indicating the probability of the input being a real HR image.

4.0.2 Loss Curve

The training and validation losses for the SRCNN model show a rapid decrease in the initial epochs and then stabilize, indicating convergence.

The generator loss shows fluctuations, which is common in GAN training, while the discriminator loss also varies as it learns to distinguish real from fake images. The validation MSE loss for SRGAN generally decreased and remained stable.

4.0.3 Metrics

Both SRCNN and SRGAN models were evaluated on a dedicated test set using quantitative metrics (PSNR, SSIM) and qualitative visual comparisons.

Quantitative Metrics Metric SRCNN SRGAN PSNR (dB) 35.91 36.07 SSIM 0.8754 0.8779

SRGAN has better PSNR: 36.07 dB vs 35.91 dB

SRGAN has better SSIM: 0.8779 vs 0.8754

The SRGAN model achieved slightly better PSNR and SSIM scores compared to the SRCNN model on this dataset.

- **Peak Signal-to-Noise Ratio (PSNR):** A quantitative measure of image reconstruction quality. Higher PSNR values indicate better reconstruction fidelity. It's often criticized for not correlating well with human perception but is a widely used objective metric.

$$PSNR = 10 \cdot \log_{10}\left(\frac{MAX_I^2}{MSE}\right) \tag{1}$$

  where MAX_I is the maximum possible pixel value of the image (1.0 for normalized images).

- **Structural Similarity Index (SSIM):** Measures the similarity between two images, taking into account luminance, contrast, and structural information. SSIM is generally considered to correlate better with human perception than PSNR. Values range from -1 to 1, with 1 indicating perfect similarity.

- **Qualitative Evaluation:**

SRCNN Visual Comparison: The SRCNN output is visually improved compared to the bicubic upsampled image, but may still exhibit some blurriness. The error map highlights areas where the reconstruction deviates most from the ground truth.

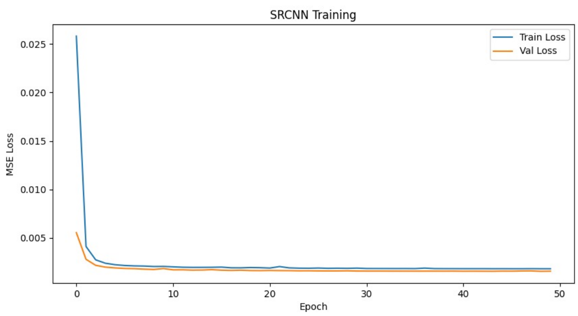
SRGAN Visual Comparison: The SRGAN output generally produces sharper and more perceptually pleasing results due to the adversarial and perceptual losses. The error maps might show a different distribution of errors compared to SRCNN, often with errors concentrated around edges and textures, reflecting its ability to generate realistic details.

4.0.4 Experiments

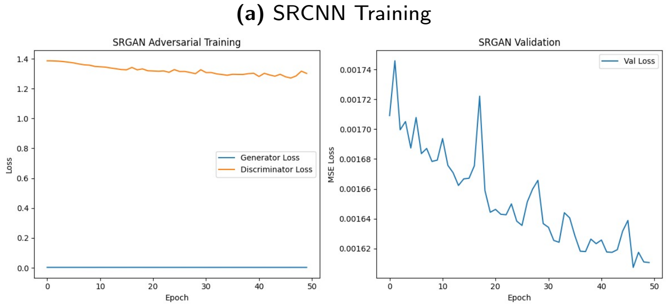
Baseline: Bicubic Interpolation As a baseline, bicubic interpolation is used to upsample the low-resolution images. This method serves as a fundamental comparison point to demonstrate the improvements offered by deep learning models. It is also the initial step for preparing inputs to both SRCNN and SRGAN in this setup.

SRCNN Training

- **Objective:** To minimize the pixel-wise MSE loss between the SRCNN output and the ground-truth HR image.
- **Optimizer:** Adam optimizer with an initial learning rate of 1e-4.
- **Learning Rate Scheduler:** StepLR with step_size=30 and gamma=0.5.
- **Epochs:** 50
- **Observations:** The SRCNN model was trained, and its training and validation losses were tracked. The loss curves demonstrate a rapid decrease in MSE loss during the initial epochs, stabilizing thereafter.



(a) SRCNN Training



(b) SRGAN Training

Figure 2: Training graphs

SRGAN Training

- **Pre-training (Generator):** The SRGAN Generator was first pre-trained for 20 epochs using only MSE loss. This helps the generator learn a basic mapping and prevents unstable training in the subsequent adversarial phase.
- **Adversarial Training:**
- **Epochs:** 50
- **Optimizers:** Adam optimizer for both generator and discriminator, with LR=1e-4 and betas=(0.9, 0.999).
- **Losses:** The generator was trained with a combined perceptual, adversarial, and MSE loss. The discriminator was trained with binary cross-entropy loss.
- **Observations:** The training process for GANs can be more challenging due to the adversarial nature. The generator and discriminator losses were monitored, along with the validation MSE loss for the generator.

Evaluation Setup Both trained models (SRCNN and SRGAN Generator) were evaluated on a held-out test set. For each test image:

- The LR image was bicubically upsampled.
- This upsampled LR image was fed to the SRCNN and SRGAN Generator.
- PSNR and SSIM were calculated between the model's output and the HR ground truth.
- Visual comparisons were generated, displaying the bicubic LR, SRCNN output, SRGAN output, and the HR ground truth, along with error maps.

5 Conclusion

This project successfully implemented and evaluated two deep learning models, SRCNN and SRGAN, for single-image super-resolution on the DIV2K dataset. The objective was to reconstruct high-resolution images from their low-resolution counterparts.

The **SRCNN**, a foundational CNN architecture for SR, demonstrated its capability to significantly improve image quality over traditional bicubic interpolation, as reflected by its PSNR and SSIM scores. Its training focused on minimizing pixel-wise Mean Squared Error, leading to quantitatively good but sometimes overly smooth results.

The **SRGAN**, built upon a generative adversarial framework with a deep residual generator and a discriminator, aimed to produce perceptually more realistic images. Through a combined loss function incorporating adversarial, perceptual (VGG-based), and pixel-wise MSE components, SRGAN managed to generate images with sharper details and textures. This was quantitatively validated by **SRGAN achieving slightly higher PSNR (36.07 dB vs 35.91 dB) and SSIM (0.8779 vs 0.8754) scores than SRCNN** on the test set, indicating its superior performance in this comparative study. Qualitatively, SRGAN's outputs often appeared more visually convincing due to its ability to hallucinate realistic high-frequency content, a known advantage of perceptual losses.

In conclusion, while SRCNN provides a solid baseline for CNN-based super-resolution, **SRGAN emerges as the better performer in this setup**, offering improved quantitative metrics and enhanced perceptual quality. The project underscores the effectiveness of deep learning, particularly adversarial networks and perceptual losses, in advancing the field of image super-resolution.

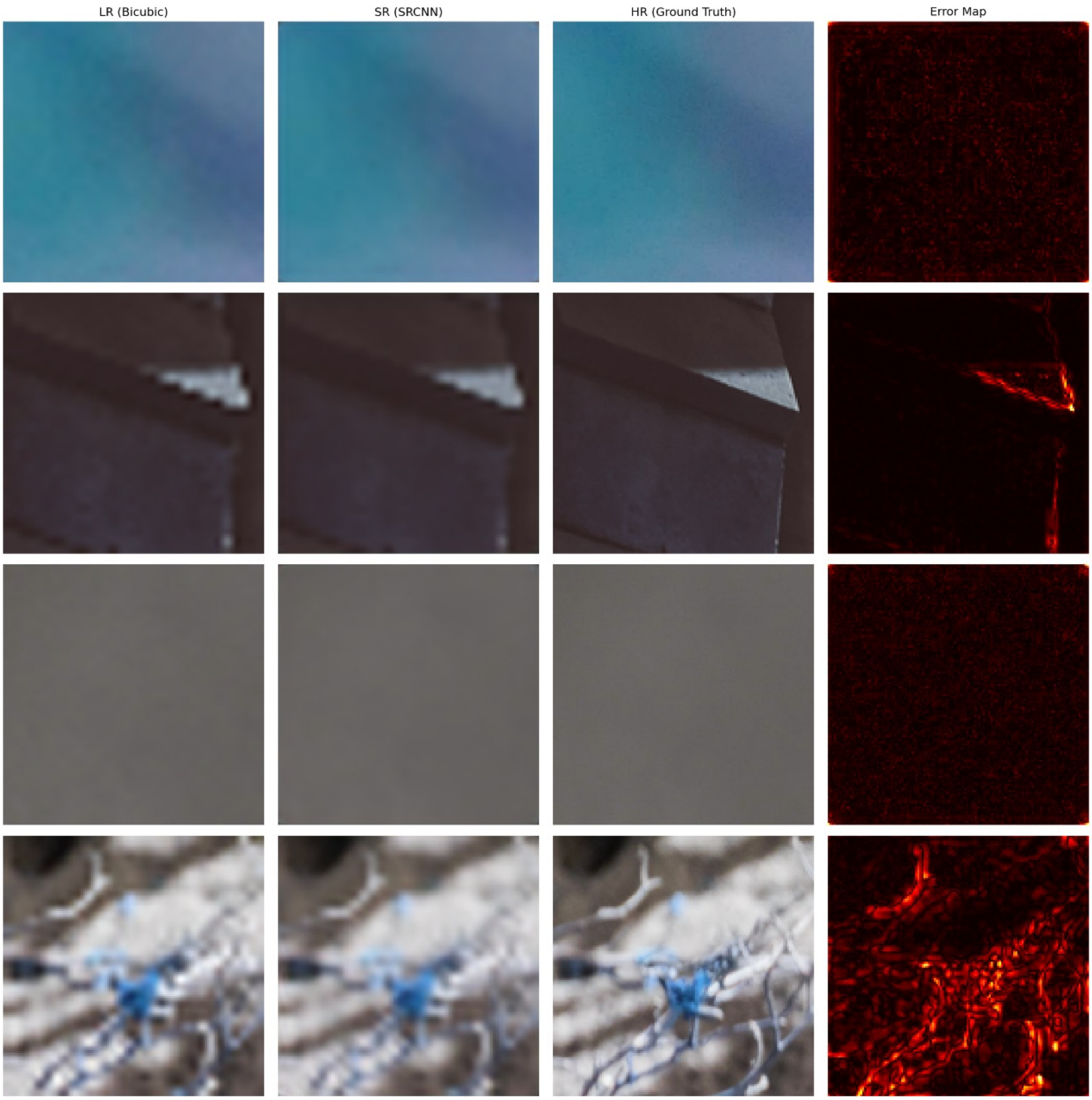


Figure 3: SRCNN

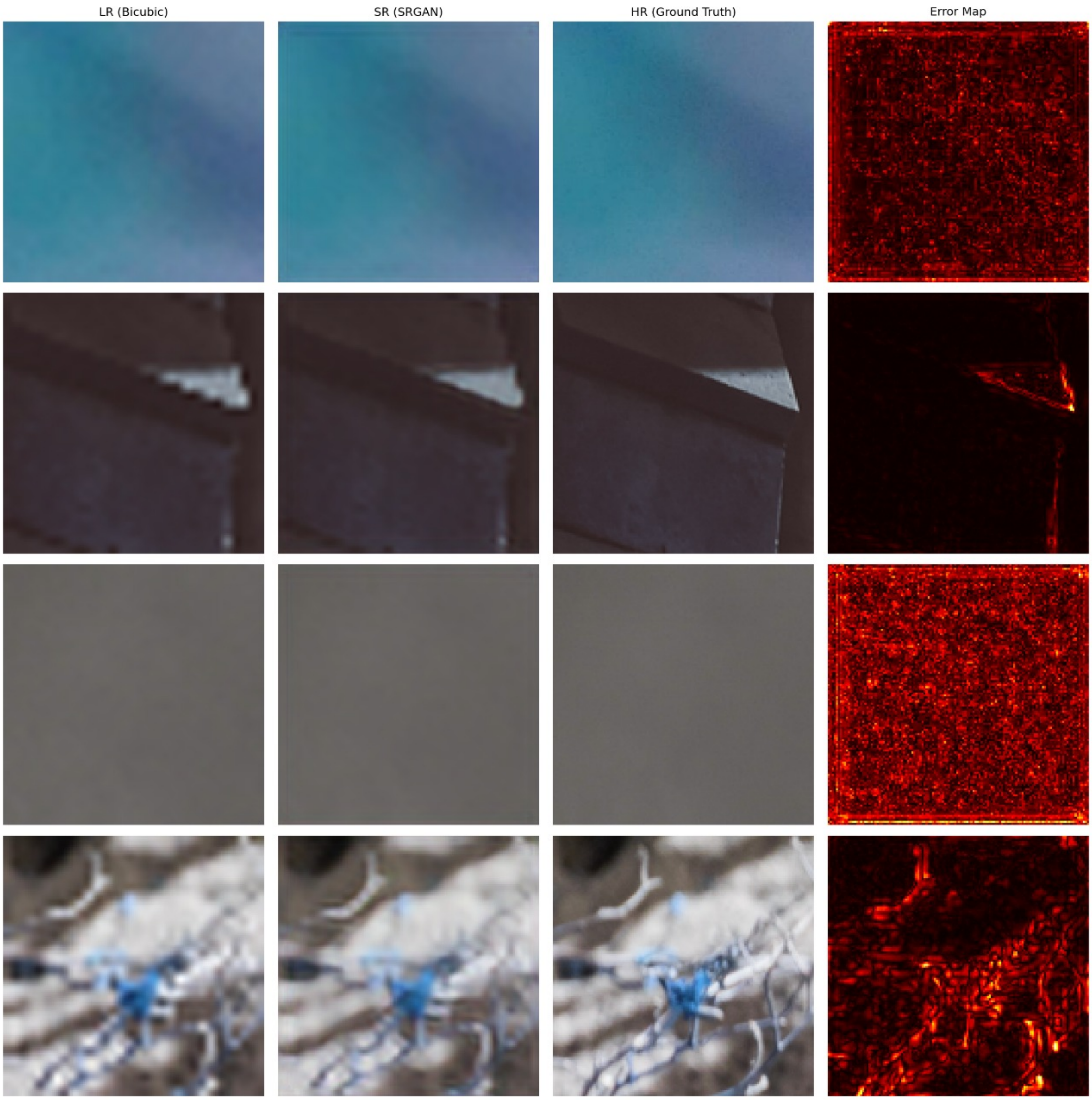


Figure 4: SRGAN